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Project Title:	All -Optical Atomic Bose Condensates		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
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Space Biology Element:	None		
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
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Task Description:	We propose an experimental program to investigate atomic Bose-Einstein condensates created using all-optical trapping methods. We will exploit the considerable flexibility provided by tailoring the optical trapping potential to explore alternative trapping geometries, allowing for for-mation of multiple condensates. This capability, combined with spin-independent nature of these traps, will allow us to explore many interesting fundamental issues of multi-component conden-sates. Our studies will include detailed examinations of the remarkable loading properties of these traps. The objective of our program is to investigate Bose-Einstein condensates formed by all-optical techniques. We will focus on several unique aspects of our system, including formation dynamics of our multi-component condensates, loading dynamics of our traps, and creation and study of condensates in multiple traps. In particular, we will further develop our all-optical BEC approach, endeavoring to create larger condensates by		

	tailoring the optical potentials. Additionally, we will apply these techniques to the study of multicomponent (ferromagnetic) spinor condensates and interacting condensates in large period optical lattices. Atomic Bose condensates may form the basis for next generation atomic clocks in space. Space-based ultracold atomic experiments may benefit significantly from the developments resulting from our program, particularly with regards to the speed and simplicity of the condensate formation together with the low power consumption of our trapping lasers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The technologies of ultracold atoms have applications to precision navigational sensors, atomic clocks for precision time metrology and quantum computing and information. Additionally, quantum degenerate gases provide a new avenue for experimental investigations of quantum many-body systems which are ubiquitous through physics.
Task Progress:	This research utilizes spinor condensates created directly in all-optical traps with which we studied the nature of the spinor ground state of the system and dynamical evolution of excited spin states. An important outcome of this work was the demonstration of coherent evolution of the spinor system, which provided the first convincing validation of the mean-field theory of spin-1 condensates. Additionally, we showed that the F = 1 Rb-87 spinor ground state was ferromagnetic in nature, confirming theoretical predictions. Multi-component condensate systems provide unique opportunities for investigating coupled, interacting quantum many-body systems and represent an active frontier in BEC research. In ultra-cold gases, multi-component systems can be realized quite simply by trapping multiple species or isotopes in the same trap. The focus of our research, however, is on Bose condensates with multiple internal states provided by the spin degrees of freedom of the atoms. These so-called spinor condensates are unique in the fact that on the one hand, they are generally capable of exploring the full range of phenomena accessible to (non-spinor) mixtures of distinguishable condensates including phase separation, modulation instability and domain formation (effects driven by the differential quantum pressures of the different quantum fluids). Beyond this however, spinor condensates offer qualitatively new phenomena deriving from the vector order parameter. For example, for spinor condensates with spin F = 1, the spin components are capable of coherently inter-converting in a process called spin-mixing due to spin-dependent collisional interactions. The focus of our initial investigations was on determining the sign of the spin-dependent collisional interaction energy, c_2, for the F = 1 Rb-87 spinor, which had been predicted to be negative, based on determinations of the scattering lengths using photoassociative spectroscopic and Feshbach resonance data. In order to experimentally verify this prediction, we adapted the method, pioneered by the MIT group, in which the spinor is created in a well-defined, non-equilibrium spin state, and then allowed to relax via spin-mixing to a final state. Recently, after improving our system and adding control over the initial state preparation, we have been able to clearly observe coherent and reversible spin dynamics in the condensates. This provides the first demonstration of the phase coherence of the spin-1 spinor and thus unequivocally validates the mean-field theoretical treatment of the system outlined above. The observed spin-mixing is an internal state analogue to Josephson oscillations in weakly connected superconductors and represents a type of matter-wave four-wave mixing. We have measured the spin oscillation frequency for different initial spin populations. These data show good agreement with the theoretical prediction and provide a direct determination of the magnitude of the spin interaction energy. Our experiment provide the first direct measurement of this important quantity, and our value agrees with the theoretical estimation using photoassociative spectroscopic and Feshbach resonance data. The large amplitude oscillations we initially observed are in the nonlinear regime. It is also possible to access the linear regime more typical of the standard Josephson effect by tuning the parameters of the system. In particular, for large applied magnetic fields s and appropriate initial populations, the phase evolution is dominated by the quadratic Zeeman effect. For these conditions, the system exhibits small oscillations analogous to AC-Josephson oscillations. We have observed these oscillations for different applied magnetic fields. Up to 12 fast oscillations are observed at the highest fields that were studied. Beyond controlling the system via the initial conditions, the dynamical evolution of the system can be controlled in real time by manipulating the spin populations and/or changing the spinor phase. We demonstrated experimentally that we can coherently control the dynamical evolution of the spinor by applying phase shifts. In particular, we showed that we have the ability to drive the system to the ferromagnetic spinor ground state using this technique. In this experiment, an initial non-equilibrium spin configuration is created and allowed to evolve for a fraction of an oscillation. At this moment, we briefly pulse on a magnetic field to apply a phase shift to the spinor. We find that for some particular applied phase shifts, the spinor condensate is brought to its ground state following the pulse, which is evidenced by the subsequent lack of population oscillation. The spinor oscillations are observed to damp with a time constant of ~250 ms. The damping coincides with the appearance of spatial spin structures apparent in the condensates. These structures indicate that the single mode approximation (SMA) is not completely valid for our experimental conditions. The validity of the SMA generally requires that the largest dimension of the condensate be smaller than the healing length associated with the spin-dependent mean field. This condition is not satisfied in the elongated condensate used in the study of the coherent dynamics, hence, the appearance of spin structures is not surprising—they result from a complicated interplay of the internal and external dynamics that ultimately transfers the internal spin energy into spatial domain structures. In the course of these investigations, we have developed and improved our fast and simple method for creating large condensates directly in optical traps. Using just a single focused laser beam and dynamical compression of the trap during evaporation, condensates up to 300,000 atoms in size are created after 2 s of evaporative cooling. SUMMARY Research in ultra-cold atoms and molecules is an important frontier of AMO physics that has the potential to significantly impact other areas of physics and technology. Scientific applications include studies of quantum degenerate gases (i.e. Bose-Einstein condensates and degenerate Fermi gases), low-energy collisions, and precision measurements of fundamental constants and symmetries. Technological applications of these systems include precision sensors for navigation and magnetometry, atomic clocks, and emerging quantum technologies including quantum information and communication.

	As discussed above, our recent research has provided the first quantitative validation of the mean-field theory of spin-1 condensates both within and outside of the single mode approximation. This work paves the way to extend spinor investigations into the non-classical regime, which will probe the relatively unexplored system size scale of <1000 atoms where complete quantum calculations are not possible in general and mean-field theoretical approaches break down. In this regime, quantum effects such as spin squeezing and entanglement become evident. Spin squeezing and spin entanglement are important themes in ultra-cold atomic physics due to potential applications in precision metrology, and investigations of these phenomena will further our basic understanding of large interacting quantum spin systems.
Bibliography Type:	Description: (Last Updated: 01/23/2008)
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Significant Media Coverage	Bigelow N. "Bose–Einstein condensates: Spins mixed up. Article highlighting research reported in 'Coherent spinor dynamics in a spin-1 Bose condensate,' in this same issue." Nature Physics 2005 Nov;1(2):89-90. http://dx.doi.org/10.1038/nphys163 , Nov-2005